# Moments-EEG: A Large-Scale EEG Dataset of Naturalistic Audiovisual Event Perception

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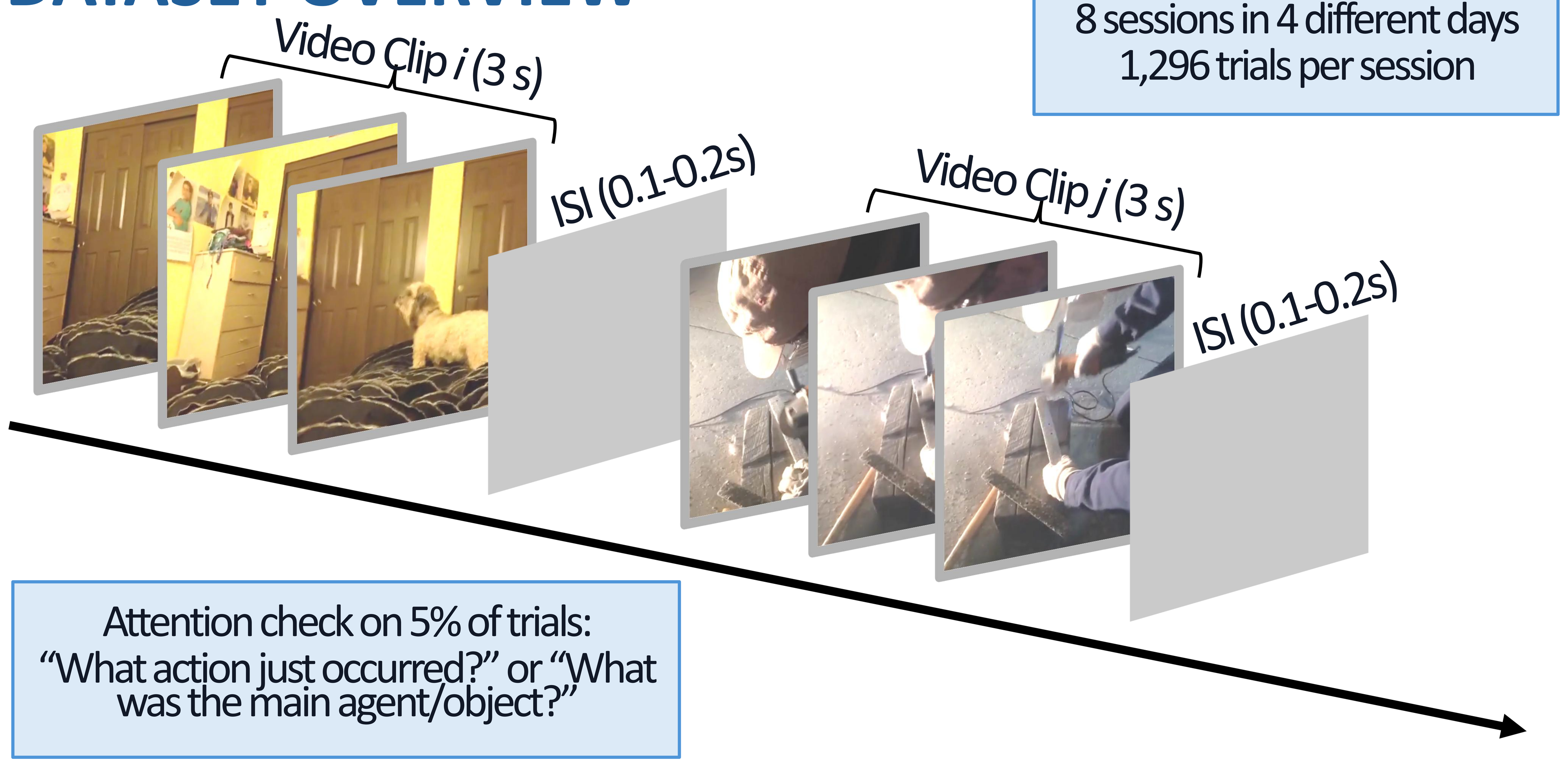
## INTRODUCTION

- Human event perception unfolds dynamically and integrates visual and auditory information.
- Understanding these processes requires high-temporal-resolution neural data under naturalistic stimulation.
- Existing EEG datasets[1,2,3] often focus on static images or simplified stimuli, limiting real-world event perception research.
- Moments-EEG provides a large-scale benchmark for temporal dynamics, multisensory processing, video-to-EEG encoding, EEG-to-video decoding, and brain-modal alignment.

## DATASET OVERVIEW

Each participant:
8 sessions in 4 different days
1,296 trials per session




Attention check on 5% of trials:
"What action just occurred?" or "What was the main agent/object?"

**Stimulus Conditions** (*Randomized video order in expt):

**16 event types:** barking, diving, drumming, frying, giggling, howling, inflating, mowing, pouring, raining, sanding, shredding, sneezing, tapping, vacuuming, whistling

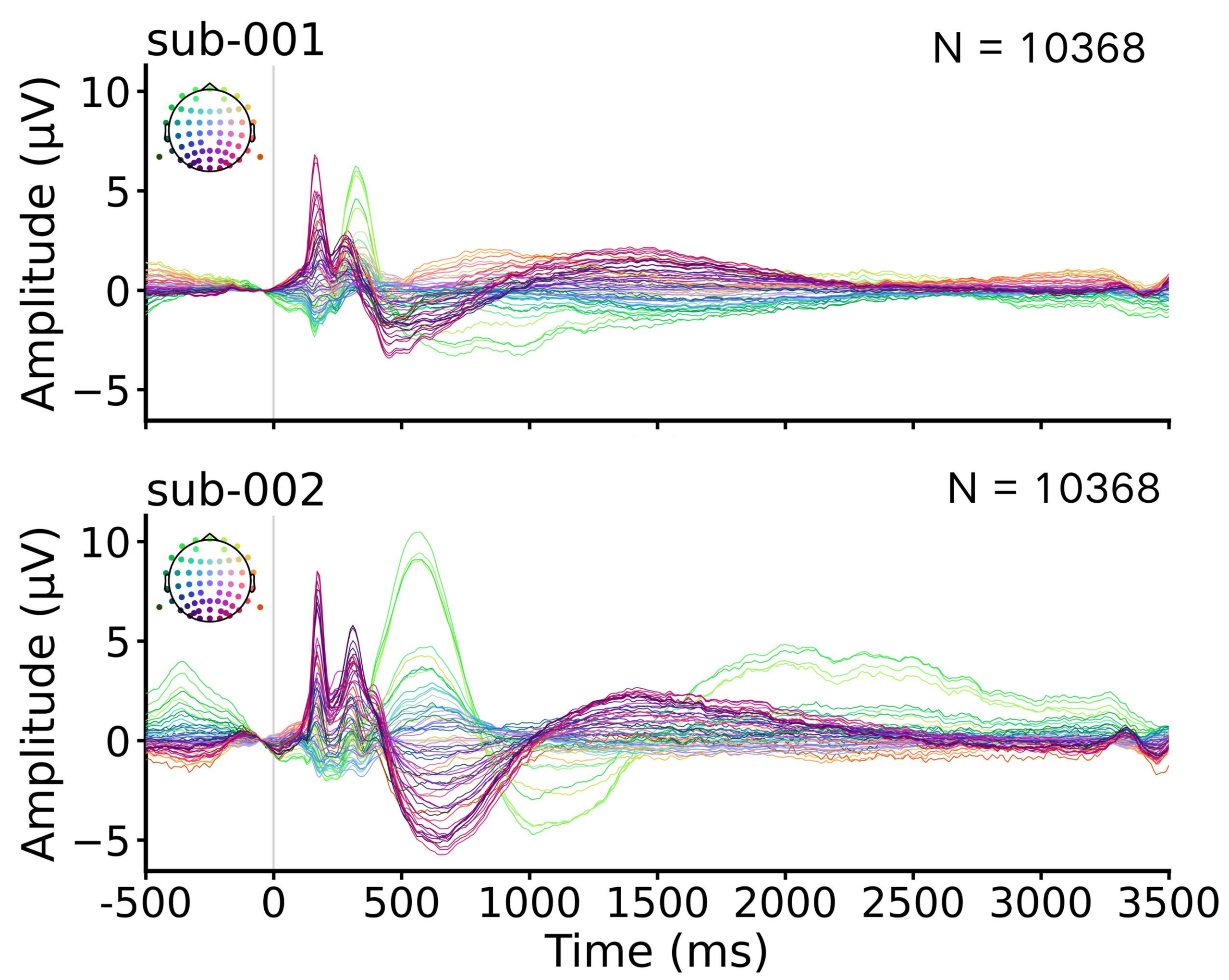
**Training Set** (896 * 3 = 2,688 trials):
intact audiovisual videos, 896 videos (56 videos per event type), 3 repetitions per video

**Test Set** (3 * 64 * 40 = 7,680 trials):
64 videos (4 videos per event type), 40 repetitions per video
Condition 1: intact audiovisual videos
Condition 2: intact visuals paired with scrambled meaningless audio
Condition 3: scrambled meaningless videos with intact audio

>= 20 human participants | 64-channel scalp EEG | 3-second naturalistic videos from *Audiovisual Moments in Time*[4] spanning 16 event types | 896 train videos + 64 test videos | 10,368 trials per participant

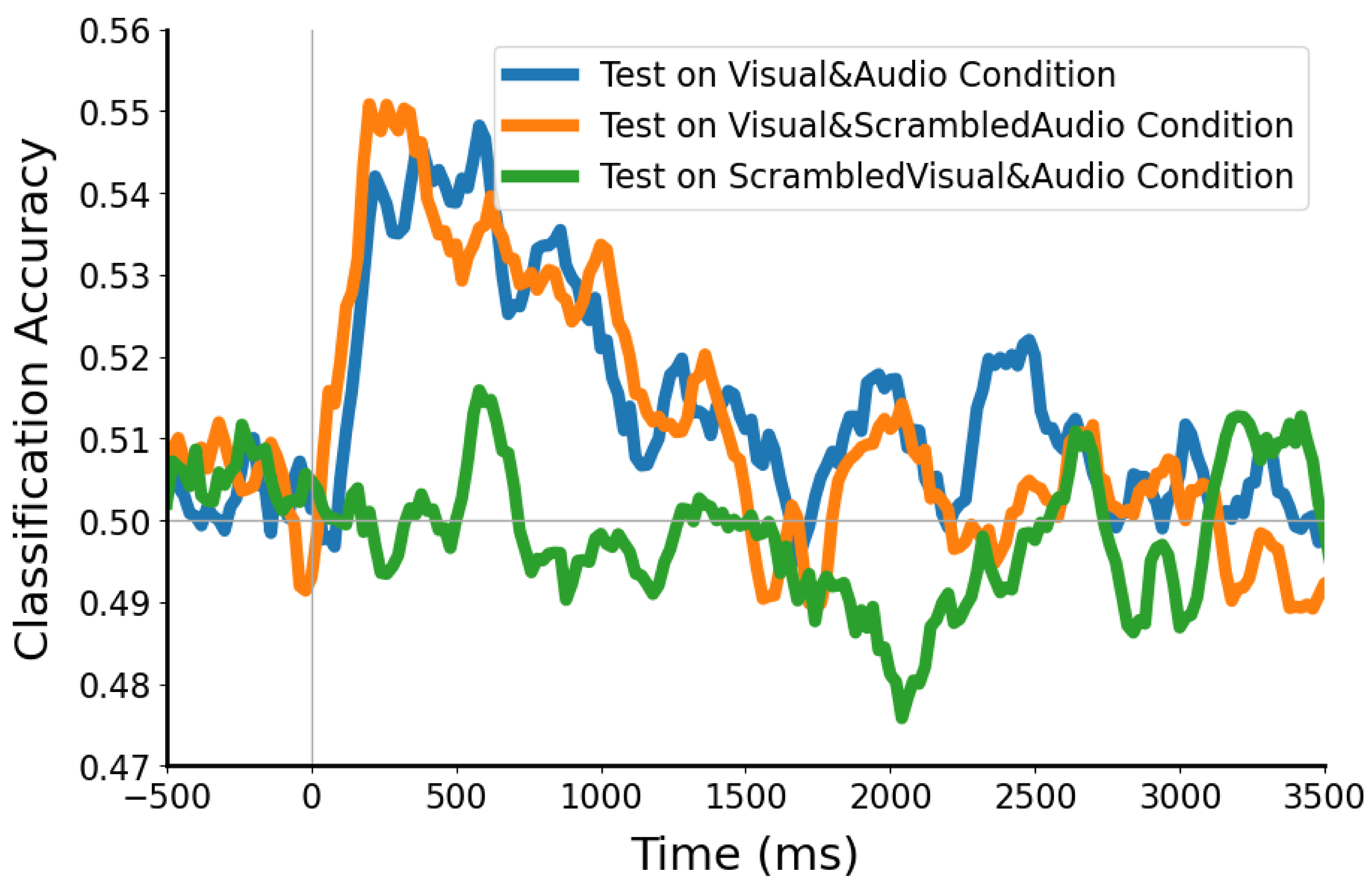
## PRELIMINARY RESULTS
**(current N = 5; ongoing data collection)**

**Preprocessed Event-Related Potentials (ERPs)**




**Classification-based EEG Decoding for 16 Event Types**

***(pair-wise decoding, trained on training data then test on test data under three different conditions)***




## POTENTIAL APPLICATIONS

- Benchmark for EEG-Video Models
- Multisensory Integration & Modality Contributions
- Neural Dynamics of Event Perception
- Model-Brain Alignment for EEG & Video Inputs
- Video-to-EEG Brain Encoding Models
- EEG-to-Video Brain Decoding Models
- Broad Impacts in Cognitive Neuroscience & NeuroAI

## AVAILABILITY / CONTACT

Data collection is currently ongoing! We expect to release the full dataset, analysis code, and a preprint describing the dataset in the second half of this year.

For questions or updates, please feel free to contact us at zitonglu@mit.edu and dongweili@bnu.edu.cn

References:
1. Spampinato et al. 2017, *CVPR*.
2. Grootswagers et al. 2022, *Scientific Data*.
3. Gifford et al. 202, *NeuroImage*.
4. Joannou et al. 2024, *PLoS One*.